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TASK	WINDOWS FORENSICS 1 REPORT

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WINDOWS FORENSICS 1 REPORT

Introduction

In this module, I explored the fundamentals of Windows digital forensics, focusing on identifying and analyzing key system artifacts. I learned how to use forensic tools to investigate user activity, system changes, and potential security incidents.

1.0 Introduction to Windows Forensics

I learned that computer forensics involves collecting and analyzing digital evidence to understand user activity on a system. I understood that Windows systems store forensic artifacts, which are traces of user actions, in various locations like the registry and user directories. I also realized that these artifacts are created to improve user experience but can be used by investigators to reconstruct events.

Question: What is the most used Desktop Operating System right now?

Answer: **Microsoft Windows**

2.0 Windows Registry and Forensics

I learned that the Windows Registry is a structured database that stores configuration data about the system, users, and applications. I understood that it is organized into keys and values, with five main root keys that help categorize different types of information. I also learned that the registry is a valuable source of forensic artifacts used to trace user activity on a system.

Question: What is the short form for HKEY_LOCAL_MACHINE?

Answer: **HKLM**

3.0 Accessing registry hives offline

I learned how to access Windows registry hives from an offline system by locating their files on disk. I understood the importance of key hives like SAM, SYSTEM, and NTUSER.DAT, as well as specialized ones like Amcache for tracking executed programs. I also learned that transaction logs and backups can provide additional forensic evidence, including recent or previously deleted data.

Question1: What is the path for the five main registry hives, DEFAULT, SAM, SECURITY, SOFTWARE, and SYSTEM?

Answer: **C:\Windows\System32\Config**

Question2: What is the path for the AmCache hive?

Answer: **C:\Windows\AppCompat\Programs\Amcache.hve**

4.0 Data acquisition

I learned that data acquisition is the process of creating a copy of system data to ensure forensic analysis does not alter the original evidence. I understood that registry hives are protected files and require specialized tools like KAPE, Autopsy, or FTK Imager to extract them. I also learned that different tools provide various methods for acquiring registry data from live systems or disk images.

5.0 Exploring Windows Registry

I learned that exported registry hives cannot be viewed using the default Windows Registry Editor and require specialized forensic tools. I understood how tools like Registry Explorer, Registry Viewer, and RegRipper are used to analyze registry data, each with different capabilities. I also learned the importance of incorporating transaction logs to ensure more accurate forensic analysis.

6.0 System Information and System Accounts

I learned how to locate important system and user information within the Windows Registry for forensic analysis. I understood how to identify details such as the OS version, active control set, computer name, time zone, network activity, and autostart programs. I also learned how the SAM hive provides detailed information about user accounts and login activity on the system.

Question: What is the Current Build Number of the machine whose data is being investigated?

Answer: **19044**

(SOFTWARE\Microsoft\Windows NT\CurrentVersion)

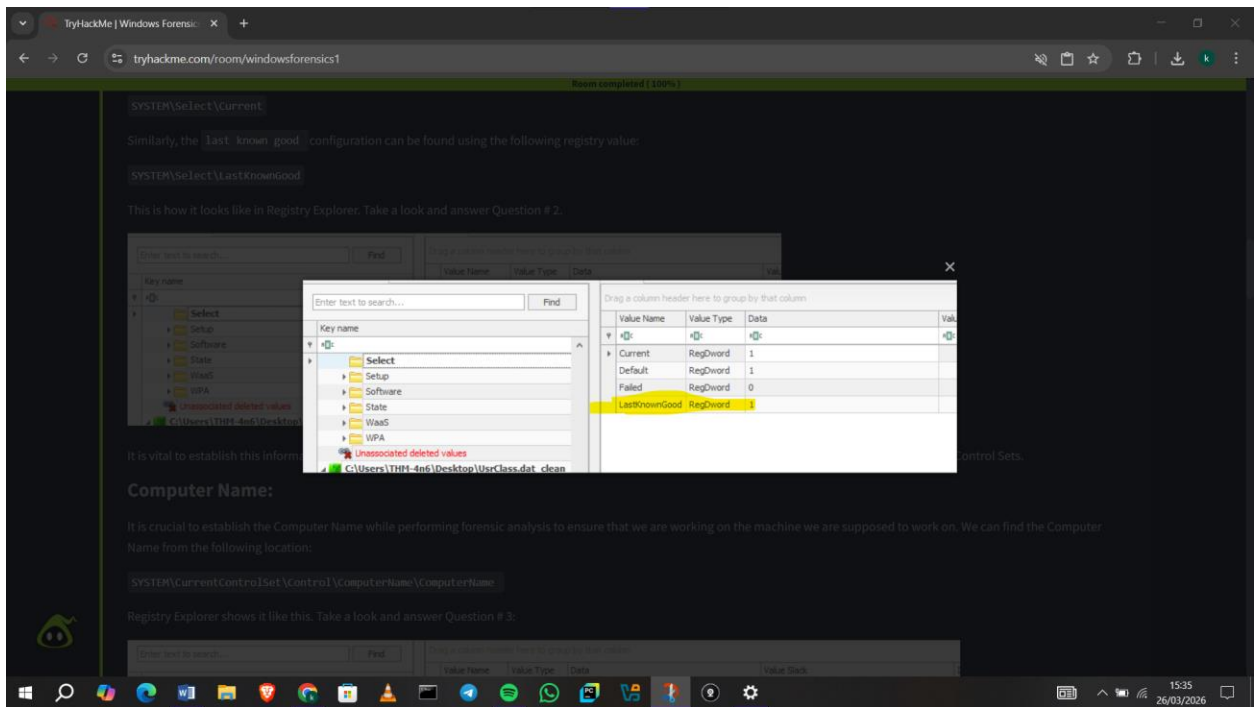
The screenshot shows the Windows Registry Editor with the path `SOFTWARE\Microsoft\Windows NT\CurrentVersion` selected. The right pane displays a list of registry values. The 'CurrentBuild' value is highlighted with a yellow box, showing the value 19044.

Value Name	Value Type	Data	Value Slack
SystemRoot	RegSz	C:\WINDOWS	00-00-00-00-00-00
BaseBuildRevisionNumber	RegDword	1	
BuildBranch	RegSz	vb_release	00-00-00-00-00-00
BuildGUID	RegSz	#####-###-###-#####	00-00
BuildLab	RegSz	19041.vb_release.191206-1406	00-00
BuildLabEx	RegSz	19041.L.ams54fe.vb_release.191206-1...	00-00-00-00
CompositionEditionID	RegSz	Enterprise	00-00-00-00-00-05
CurrentBuild	RegSz	19044	
CurrentBuildNumber	RegSz	19044	
CurrentMajorVersionNumber	RegDword	10	
CurrentMinorVersionNumber	RegDword	0	
CurrentType	RegSz	Multiprocessor Free	65-00-64-00-00-00-00-00-00-00-00-00
CurrentVersion	RegSz	6.3	00-00-00-00
EditionID	RegSz	Professional	00-00
EditionSubManufacturer	RegSz		
EditionSubstring	RegSz		
EditionSubVersion	RegSz		
InstallationType	RegSz	Client	00-00-00-00-00-00
InstallDate	RegDword	1637778211	
ProductName	RegSz	Windows 10 Pro	72-00-70-00-72-00-69-00-73-00-65-00-0...
ReleaseId	RegSz	2009	00-00
SoftwareType	RegSz	System	00-00-00-00-00-00

Question: Which ControlSet contains the last known good configuration?

Answer: **1**

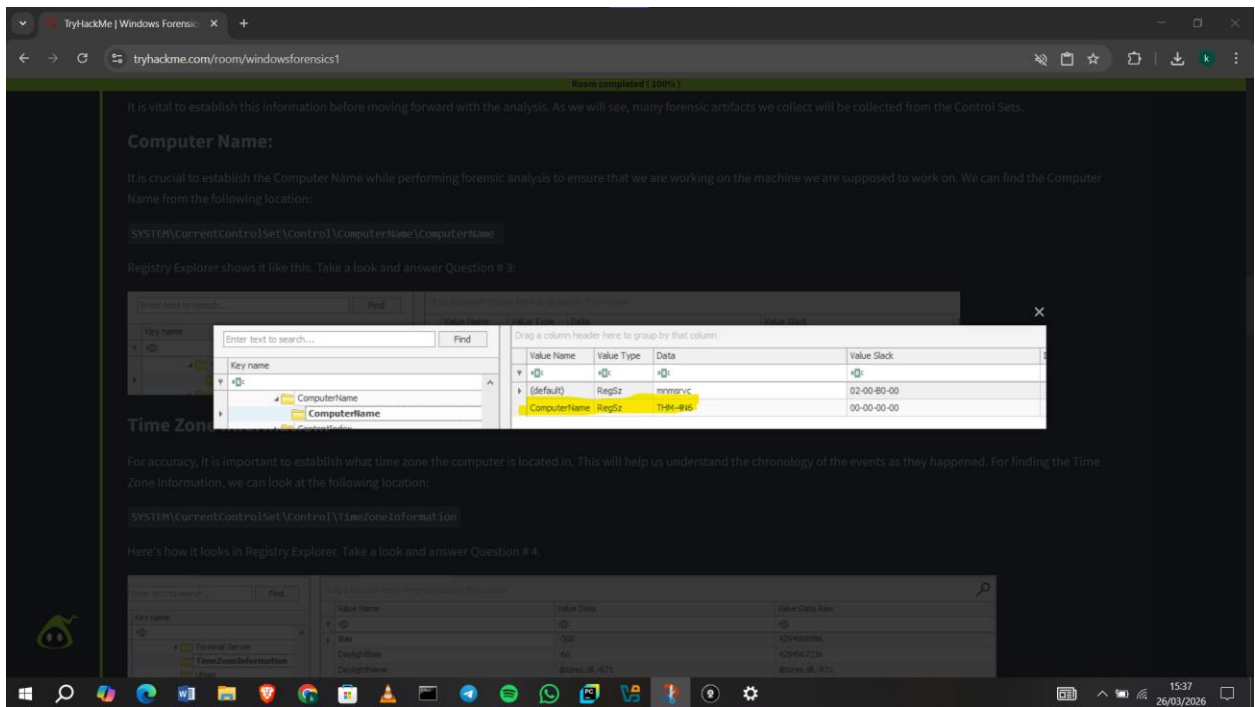
(SYSTEM\Select\LastKnownGood)



Question: What is the Computer Name of the computer?

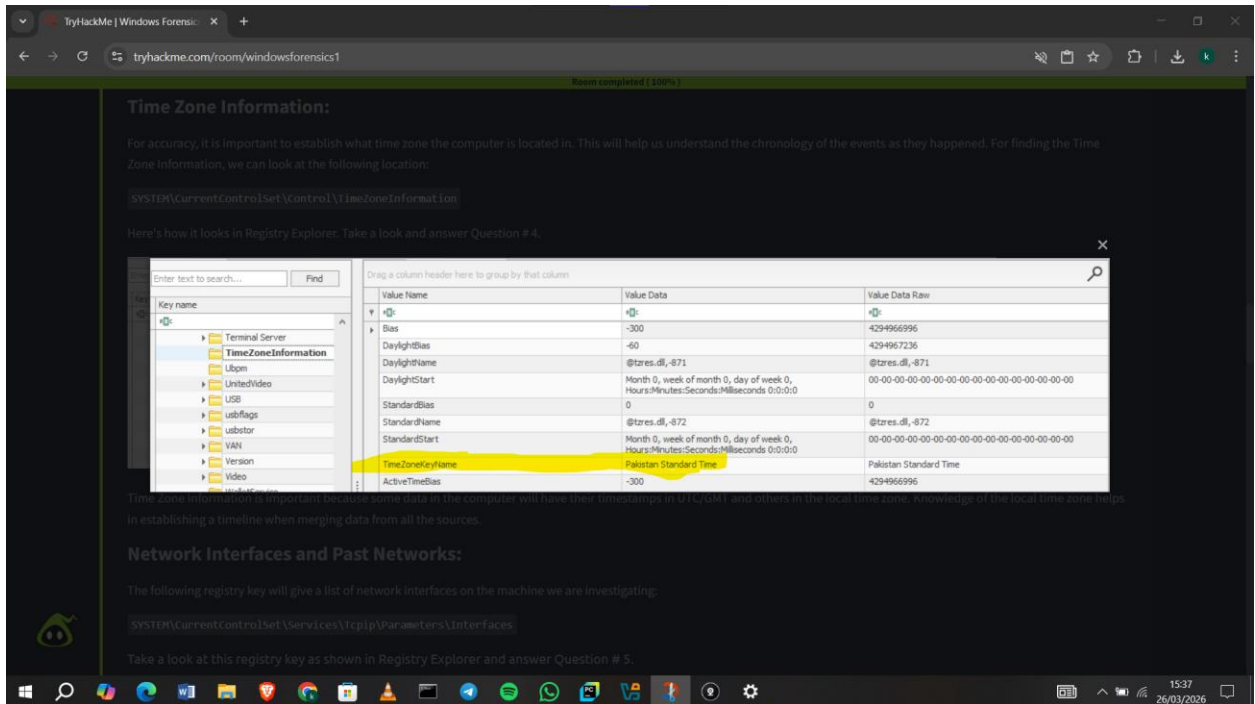
Answer: **THM-4n6**

(SYSTEM\CurrentControlSet\Control\ComputerName\ComputerName)



Question: What is the value of the TimeZoneKeyName?

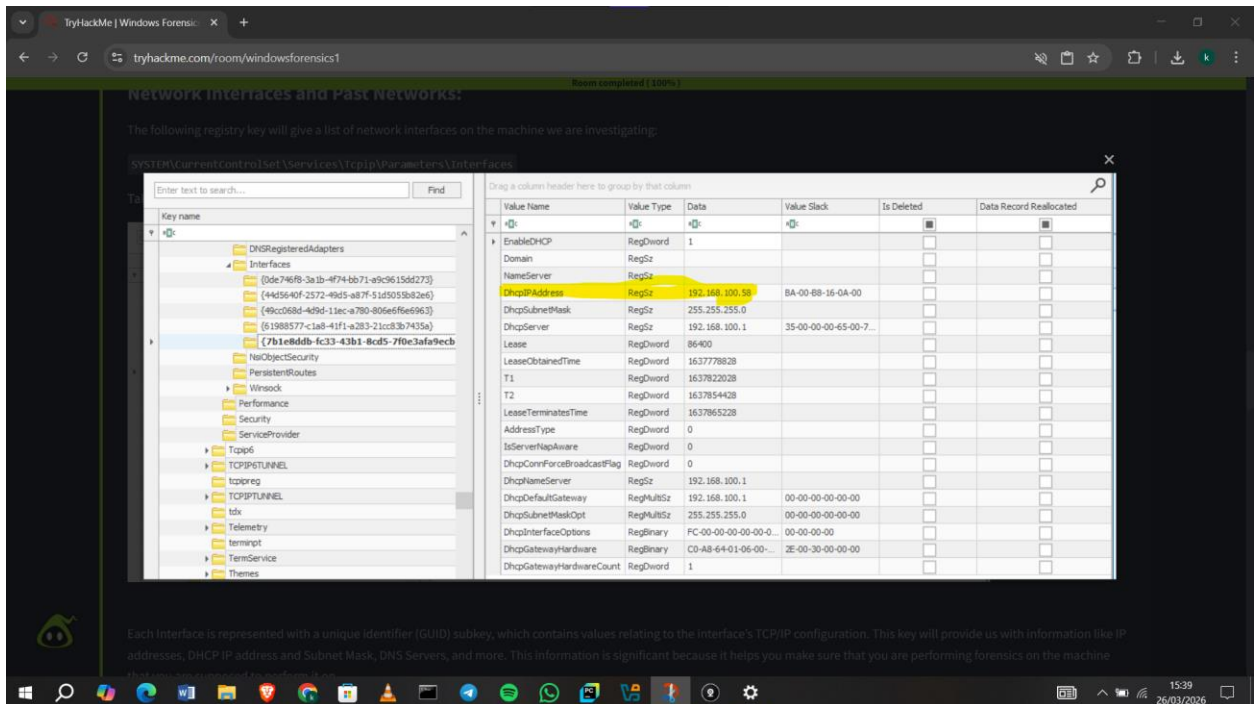
Answer: Pakistan Standard Time
(SYSTEM\CurrentControlSet\Control\TimeZoneInformation)



Question: What is the DHCP IP address?

Answer: **192.168.100.58**

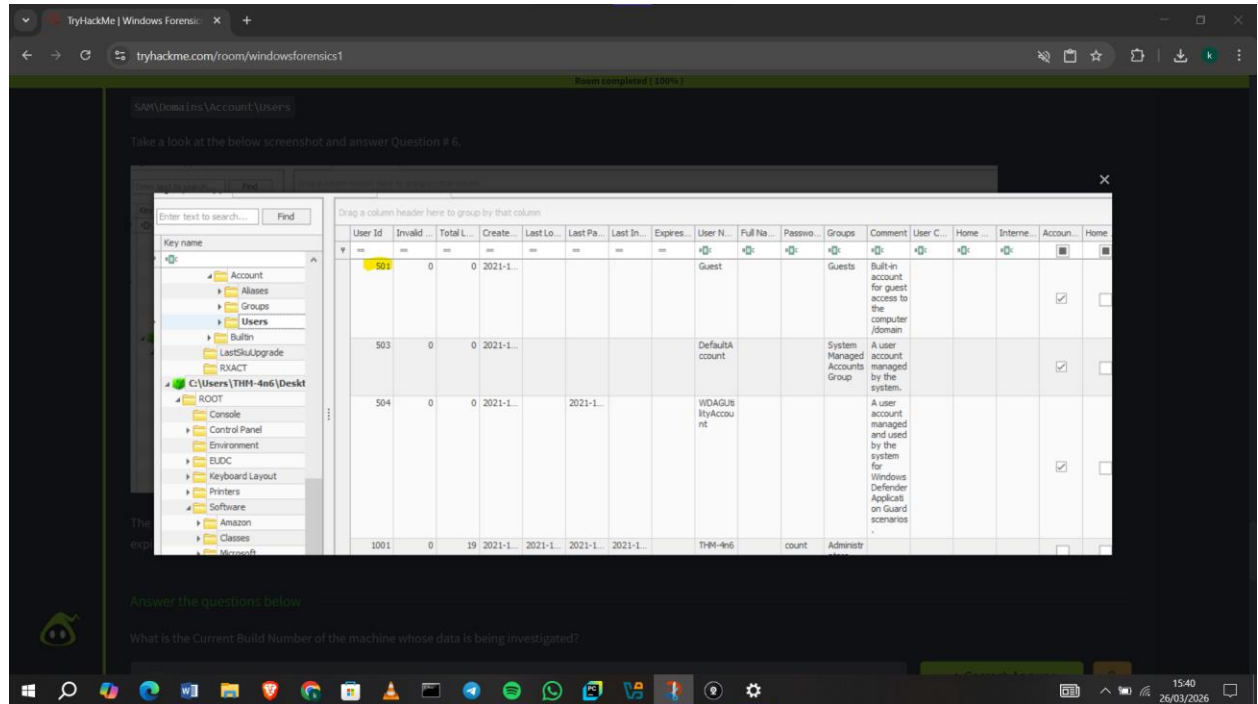
(SYSTEM\CurrentControlSet\Services\Tcpip\Parameters\Interfaces)



Question: What is the RID of the Guest User account?

Answer: **501**

(SAM\Domains\Account\Users)



7.0 Usage or knowledge of files/folders

I learned how to identify a user's interaction with files using various registry artifacts such as RecentDocs, Office recent files, and ShellBags. These artifacts can reveal recently opened files, accessed folders, and even deleted file paths. I also learned that additional registry keys can show file access through open/save dialogs and user search or navigation activity.

Question: When was EZtools opened?

Answer: **2021-12-01 13:00:34**

(NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs)

TryHackMe | Windows Forensics: X +

tryhackme.com/room/windowsforensics1

Room completed (100%)

Recent Files:

Windows maintains a list of recently opened files for each user. As we might have seen when using Windows Explorer, it shows us a list of recently used files. This information is stored in the NTUSER hive and can be found on the following location:

Registry Hives (7) Available bookmarks (108/0)

Enter text to search... Find

Recent documents

Drag a column header here to group by that column

Extension	Value Name	Target Name	Link Name	Mru Position	Opened On	Extension Last Opened
RecentDocs	7	EZtools	EZtools.lnk	0	2021-12-01 13:06:34	
RecentDocs	6	Settings	Settings.lnk	1		2021-11-30 10:56:23
RecentDocs	5	WallpaperSettings.xml	WallpaperSettings.lnk	2		2021-11-30 10:56:21
RecentDocs	4	System and Security	System and Security.lnk	3		
RecentDocs	3	::{B806CDE4-0293-4F75-8A9D-CB5356477EE}	System.lnk	4		
RecentDocs	1	KAPE	KAPE.lnk	5		
RecentDocs	0	Get-KAPEUpdate.ps1	Get-KAPEUpdate.lnk	6		2021-11-24 18:18:48
RecentDocs	2	ChangeLog.txt	ChangeLog.lnk	7		2021-11-24 18:18:48
Folder	2	Settings	Settings.lnk	0	2021-11-30 10:56:23	
Folder	1	System and Security	System and Security.lnk	1		
Folder	0	KAPE	KAPE.lnk	2		
.xml	0	WallpaperSettings.xml	WallpaperSettings.lnk	0	2021-11-30 10:56:21	
.txt	0	ChangeLog.txt	ChangeLog.lnk	0	2021-11-24 18:18:48	
.ps1	0	Get-KAPEUpdate.ps1	Get-KAPEUpdate.lnk	0	2021-11-24 18:18:48	

Registry Explorer allows us to sort data contained in registry keys quickly. For example, the Recent documents tab arranges the Most Recently Used (MRU) file at the top of the list. Registry Explorer also arranges them so that the Most Recently Used (MRU) file is shown at the top of the list and the older ones later.

Another interesting piece of information in this registry key is that there are different keys with file extensions, such as .pdf, .jpg, .docx, etc. These keys provide us with information about the last used files of a specific file extension. So if we are looking specifically for the last used PDF files, we can look at the following registry key:

1541
26/03/2026

Question: At what time was My Computer last interacted with?

Answer: **2021-12-01 13:06:47**

(USRCLASS.DAT\Local Settings\Software\Microsoft\Windows\Shell\Bags)

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tryhackme.com/room/windowsforensics1

Room completed (100%)

NTUSER.DAT\Software\Microsoft\Windows\Shell\Bags\My Computer

NTUSER.DAT\Software\Microsoft\Windows\Shell\Bags

Registry Explorer doesn't give us much information about ShellBags. However, another tool from Eric Zimmerman's tools called the ShellBag Explorer shows us the information in an easy-to-use format. We just have to point to the hive file we have extracted, and it parses the data and shows us the results. An example is shown below. Take a look and answer Question # 2.

ShellBags Explorer v1.4.0.0

File Tools Help

Value

Desktop

- My Computer
- KAPE
- Home Folder
- Search Folder
- Control Panel
- E:\

Drag a column header here to group by that column

Value	Icon	Shell Type	MRU Pos...	Created On	Modified On	Accessed On	First Interacted	Last Interacted	Has Explored	Miscellaneous
My Computer	No icon	Root folder: GUID	0					2021-12-01 13:06:47	☒	
KAPE	Directory	Root folder: GUID	1	2021-11-25 03:34:14	2021-11-25 03:34:14	2021-11-25 03:34:14			☒	NTFS file system
Home Folder		Root folder: GUID	2				2021-11-24 18:20:02		☒	
Search Folder		Users property view	3				2021-11-30 11:08:01		☒	
Search Folder		Users property view	4				2021-11-30 11:08:52		☒	
Control Panel		Root folder: GUID	5						☒	
E:\		Users property view: Drive letter	6				2021-11-24 18:20:02		☒	

NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSavePid1MRU

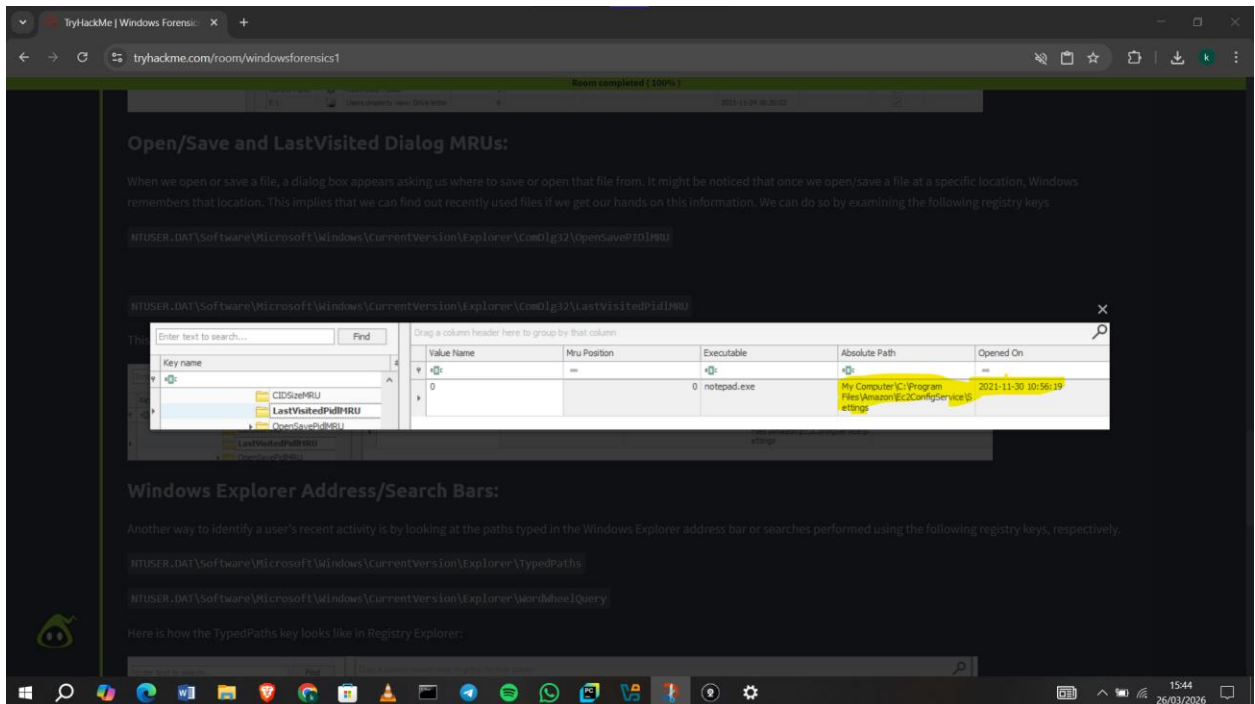
NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\LastVisitedPid1MRU

This is how Registry Explorer shows this registry key. Take a look to answer Question # 3 and 4.

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26/03/2026

Question: What is the Absolute Path of the file opened using notepad.exe?

Answer: C:\Program Files\Amazon\Ec2ConfigService\Settings
(NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\LastVisitedPidIMRU)



Question: When was this file opened?

Answer: **2021-11-30 10:56:19**

8.0 Evidence of Execution

I learned how to identify evidence of program execution using registry artifacts such as UserAssist, ShimCache, AmCache, and BAM/DAM. They provide different levels of detail, including execution times, file paths, and frequency of use. I also learned that combining multiple sources helps build a more accurate picture of application activity on a system.

Question: How many times was the File Explorer launched?

Answer: **26**

(NTUSER.DAT\Software\Microsoft\Windows\Currentversion\Explorer\UserAssist\{GUID}
\Count)

Room completed (100%)

Important for a forensic investigator, in this task, we will go through the different ways to retrieve information on connected devices and the device on a system using the registry.

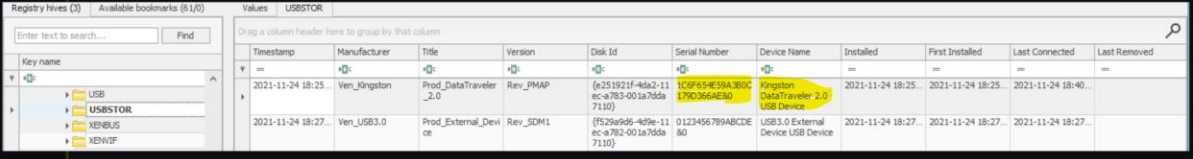
Device identification:

The following locations keep track of USB keys plugged into a system. These locations store the vendor id, product id, and version of the USB device plugged in and can be used to identify unique devices. These locations also store the time the devices were plugged into the system.

`SYSTEM\CurrentControlSet\Enum\USBSTOR`

`SYSTEM\CurrentControlSet\Enum\USB`

Registry Explorer shows this information in a nice and easy-to-understand way. Take a look at this and answer Questions # 1 and 2.



Timestamp	Manufacturer	Title	Version	Disk Id	Serial Number	Device Name	Installed	First Installed	Last Connected	Last Removed
2021-11-24 18:25...	Ven_Kingston	Prod_DataTraveler_2.0	Rev_PMAP	(e251921f-4da2-11ec-a783-001a76da7110)	(C0F654E58A30C79D366AE80)	Kingston DataTraveler 2.0 USB Device	2021-11-24 18:25...	2021-11-24 18:25...	2021-11-24 18:40...	
2021-11-24 18:27...	Ven_USB3.0	Prod_External_Device	Rev_SDM1	(f52ba9d6-4d9e-11ec-a782-001a76da7110)	0123456789ABCDE80	USB3.0 External Device USB Device	2021-11-24 18:27...	2021-11-24 18:27...	2021-11-24 18:27...	

First/Last Times:

Similarly, the following registry key tracks the first time the device was connected, the last time it was connected and the last time the device was removed from the system.

`SYSTEM\CurrentControlSet\Enum\USBSTOR\Ven_Prod_Version\USBSerial\Properties\{83da6326-97a6-4888-9453-a19231573b29}\####`

In this key, the #### sign can be replaced by the following digits to get the required information:

Value	Information
0000	First Connected
0001	Last Connected
0002	Last Removed

Question: What is the name of this device?

Answer: **Kingston Data Traveler 2.0 USB Device**

Question: What is the friendly name of the device from the manufacturer 'Kingston'?

Answer: **USB**

(SOFTWARE\Microsoft\Windows Portable Devices\Devices)

Question: What is the username of the account that has never been logged in?

Answer: I found **thm-user2** with 0 login count at **SAM\Domains\Account\Users**.

Registry Explorer v1.6.0.0

File Tools Options Bookmarks (1/0) View Help

Registry Hives (1) Available bookmarks (1/0)

Enter text to search... Find

Key name # values # subkeys Last write time

Key: SAM\Domains\Account\Users

Selected hive: SAM Last write: 2021-11-24 18:39:15 1 of 1 values shown (100.00%) Load complete

User	Inva	Tot	Crea	Last	Last	Last	Exp	User	Full	Pass	Grou	Com	User	Hom	Inte	Acco	Hom	Pass	Tem	Nor	Mns	Inte	Wor	Serv	Pass	Auto L
1001	0	19	202...	202...	202...	202...		THM-4n6	count	Administrators																
1002	0	2	202...	202...	202...	202...		thm-user	null	Users																
1003	0	0	202...	202...	202...	202...		thm-user2	null	Users																

Total rows: 7

Type viewer Binary viewer

Value name (default)

Value type RegLink

Value

Value: (default) Collapse all hives

Hidden keys: 0

2:06 PM 3/26/2026

1h 18min 33s

12:06 26/03/2026

Question: What's the password hint for the user THM-4n6?

Answer: **count**

Registry Explorer v1.6.0.0

File Tools Options Bookmarks (1/0) View Help

Registry Hives (1) Available bookmarks (1/0)

Enter text to search... Find

Key name # values # subkeys Last write time

Key: SAM\Domains\Account\Users

Selected hive: SAM Last write: 2021-11-24 18:39:15 1 of 1 values shown (100.00%) Load complete

User	Inva	Tot	Crea	Last	Last	Last	Exp	User	Full	Pass	Grou	Com	User	Hom	Inte	Acco	Hom	Pass	Tem	Nor	Mns	Inte	Wor	Serv	Pass	Auto L
1001	0	19	202...	202...	202...	202...		THM-4n6	count	Administrators																
1002	0	2	202...	202...	202...	202...		thm-user	null	Users																
1003	0	0	202...	202...	202...	202...		thm-user2	null	Users																

Total rows: 7

Type viewer Binary viewer

Value name (default)

Value type RegLink

Value

Value: (default) Collapse all hives

Hidden keys: 0

2:07 PM 3/26/2026

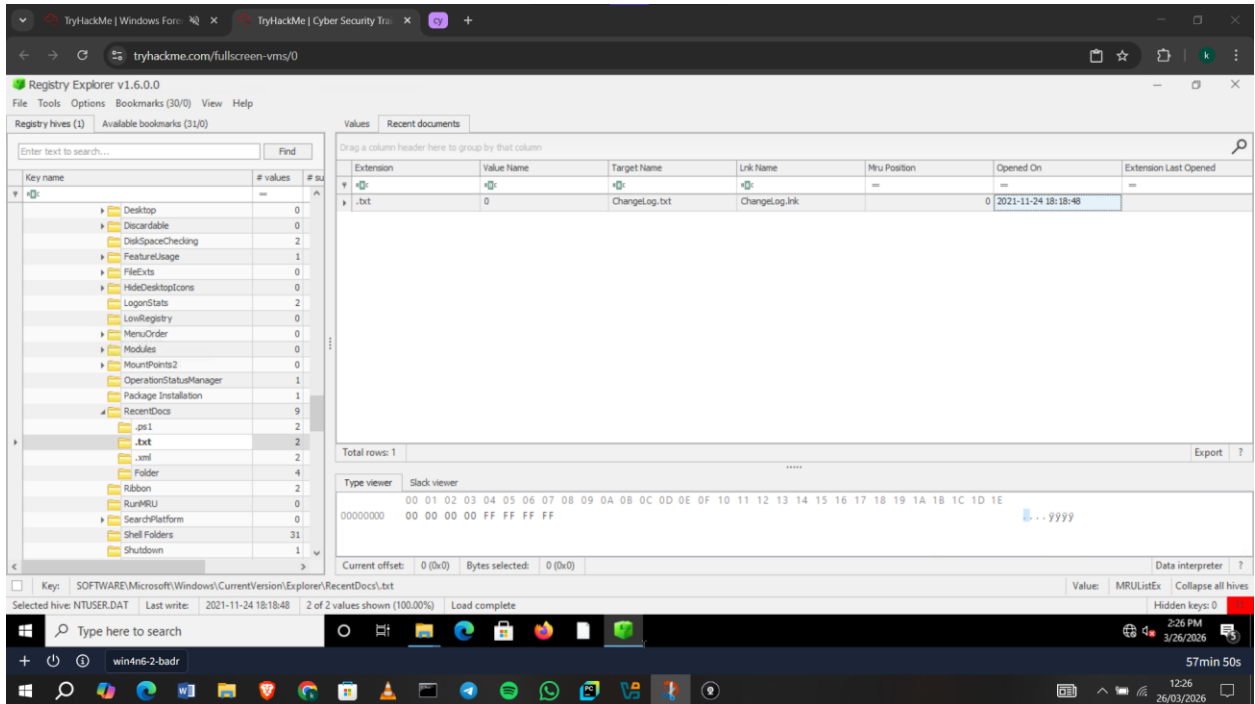
1h 17min 21s

12:07 26/03/2026

Question: When was the file 'Changelog.txt' accessed?

Answer: I found the file at

NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs and found the last date it was opened as **2021-11-24 18:18:48**.



Question: What is the complete path from where the python 3.8.2 installer was run?

Answer: I went to

NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist and found the path: **Z:\setups\python-3.8.2.exe**.

I then went to **SYSTEM\CurrentControlSet\Enum\USBSTOR** and found the Disk ID for USB and the last time it was connected was **2021-11-24 18:40:06**.

Timestamp	Manufacturer	Title	Version	Disk ID	Serial Number	Device Name	Installed	First Installed	Last Connected	Last Removed
2021-11-24 18:25:...	Ven_Kingston	Prod_DataTraveler_2.0	Rev_PMAP	{e251921f-4da2-11e1-c-a783-00 1a765a71 10}	1C6F654E59A3B0C179D366AE80	Kingston DataTraveler 2.0 USB Device	2021-11-24 18:25:...	2021-11-24 18:25:...	2021-11-24 18:40:06	
2021-11-24 18:27:...	Ven_USB3.0	Prod_External_Devic	Rev_SDM1	{752a9d6c-4d9e-11e1-c-a783-00 1a765a71 10}	0123456789ABCDEF0	USB 3.0 External Device USB Device	2021-11-24 18:27:...	2021-11-24 18:27:...	2021-11-24 18:27:...	

Great work kenmuiruri07! Room completed!

Your skills are skyrocketing!

Completed tasks: 11

Points earned: 216

Streak: 2

Scroll down to review your progress

Conclusion

Overall, this module strengthened my understanding of how digital evidence is stored and analyzed within Windows systems. I am now more confident in using forensic techniques and tools to reconstruct events and support investigations.